

# Test suite — final report

AI Supply Chain Terminal

2026-07-21

## 1. What was built

A permanent pytest suite that runs against a frozen graph snapshot in `backend/tests/fixtures/`. No live API calls, no network dependency, deterministic across runs.

```
backend/tests/
├── conftest.py           ─ session-scoped graph, config,
                        ─ narration builder fixtures
├── helpers.py           ─ pure helpers reused by tests
                        ─ + the report script
├── fixtures/            ─ frozen snapshot (no live API)
│   ├── ai/{nodes,edges,events}.json
│   ├── scoring.yaml
│   └── narration.yaml
├── test_share_completeness.py    ─ Test 1 (fails)
├── test_thin_buckets.py         ─ Test 2 (probe)
├── test_structural_no_zero.py   ─ Test 3
├── test_idempotence.py         ─ Test 4
├── test_tier_coherence.py       ─ Test 5 (+ writes schema gap)
├── test_config_coherence.py     ─ Test 6
├── test_bounds.py              ─ Test 7
├── test_graph_integrity.py      ─ Test 8
├── test_paper_chokepoints.py    ─ Test 9
├── test_narration.py           ─ Test 10
├── test_outbound_sensitivity.py ─ Test 11 (probe)
├── run_report.py               ─ bundles everything
└── _out/                      ─ one artefact per report deliverable
    ├── share_backlog.txt
    ├── thin_buckets.txt
    ├── outbound_sensitivity.txt
    └── schema_gap.txt
```

Design rule from the spec held throughout: **invariants, not values**. The only place specific values are asserted is the paper-chokepoint list in Test 9, per the spec's strict exception.

## 2. Pytest summary — 22 tests, 21 pass, 1 fail

Failing by design, not fixed:

| Test                                              | Why it fails                                                                                                                                                                                                        |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>test_no_stage_bucket_sums_below_0_80</code> | 26 stage buckets sum below 0.80. Largest offenders: single-edge <code>input_to</code> buckets at 0.08 sum (TSMC, SK Hynix, Micron, several data-centre facilities). Same nodes the thin-graph TODO already flagged. |

All 21 other tests pass — including `test_structural_no_zero` (the assertion that would have caught the previous stage-list defect), `test_every_paper_chokepoint_is_critical`, and every narration invariant.

### 3. Share-completeness backlog — 29 buckets below 0.95

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The main deliverable of this task. Sorted by shortfall, worst first.

| severity | sum   | edges | node                      | stage    |
|----------|-------|-------|---------------------------|----------|
| FAIL     | 0.080 | 1     | TSMC                      | input_to |
| FAIL     | 0.080 | 1     | SK Hynix                  | input_to |
| FAIL     | 0.080 | 1     | Micron                    | input_to |
| FAIL     | 0.080 | 1     | Colossus                  | input_to |
| FAIL     | 0.080 | 1     | Stargate (Abilene)        | input_to |
| FAIL     | 0.080 | 1     | Vantage Frontier          | input_to |
| FAIL     | 0.080 | 1     | The Citadel               | input_to |
| FAIL     | 0.090 | 2     | Samsung Electronics       | input_to |
| FAIL     | 0.250 | 1     | Amazon                    | input_to |
| FAIL     | 0.250 | 1     | Microsoft                 | input_to |
| FAIL     | 0.250 | 1     | Meta                      | input_to |
| FAIL     | 0.300 | 2     | Constellation Energy      | supplies |
| FAIL     | 0.300 | 1     | Quanta Services           | input_to |
| FAIL     | 0.450 | 2     | GE Vernova                | input_to |
| FAIL     | 0.500 | 2     | Siemens Energy            | input_to |
| FAIL     | 0.530 | 3     | Vertiv                    | input_to |
| FAIL     | 0.600 | 1     | RF & Power Semiconductors | input_to |
| FAIL     | 0.600 | 3     | Duke Energy               | supplies |
| FAIL     | 0.650 | 5     | Copper                    | mines    |
| FAIL     | 0.650 | 3     | Micron                    | supplies |
| FAIL     | 0.700 | 3     | Intel                     | supplies |
| FAIL     | 0.750 | 4     | Indium                    | mines    |
| FAIL     | 0.750 | 3     | NextEra Energy            | supplies |
| FAIL     | 0.750 | 4     | The Citadel               | supplies |
| FAIL     | 0.800 | 2     | xAI                       | supplies |
| FAIL     | 0.800 | 2     | OpenAI                    | supplies |
| warn     | 0.800 | 2     | NdFeB Permanent Magnets   | input_to |
| warn     | 0.900 | 4     | Gallium                   | refines  |
| warn     | 0.930 | 4     | Indium                    | refines  |

Same list is written to `backend/tests/_out/share_backlog.txt` .

## 4. Thin-bucket census — 14 single-edge buckets, all win the `max`

Every one of the 14 is decided by a bucket containing exactly one edge, so its HHI is 1.0 by construction and it wins `combine: max` for the node's `inbound_hhi`. This is the numeric size of the thin-graph problem.

| won max | bucket HHI | node                      | stage    |
|---------|------------|---------------------------|----------|
| yes     | 1.000      | AMD                       | supplies |
| yes     | 1.000      | Amazon                    | input_to |
| yes     | 1.000      | Broadcom                  | supplies |
| yes     | 1.000      | Colossus                  | input_to |
| yes     | 1.000      | Meta                      | input_to |
| yes     | 1.000      | Micron                    | input_to |
| yes     | 1.000      | Microsoft                 | input_to |
| yes     | 1.000      | Quanta Services           | input_to |
| yes     | 1.000      | RF & Power Semiconductors | input_to |
| yes     | 1.000      | SK Hynix                  | input_to |
| yes     | 1.000      | Stargate (Abilene)        | input_to |
| yes     | 1.000      | TSMC                      | input_to |
| yes     | 1.000      | The Citadel               | input_to |
| yes     | 1.000      | Vantage Frontier          | input_to |

Nodes whose `inbound_hhi` is decided by a single-edge bucket: **14**.

## 5. Outbound sensitivity — removing TSMC shifts 4 tiers

`outbound_criticality` normalises to graph max, so any single node's raw score can drag every other node's score. Removing the highest-outbound node (TSMC, raw 1.000) and renormalising:

```
mineral:copper      high → critical
company:kla         none → moderate
company:mp_materials none → moderate
company:lynas       none → moderate
```

**4 tiers move on the removal of one node.** When robotics/aerospace land on the same graph, AI tiers will shift for non-AI reasons proportional to this number.

## 6. Baseline-vs-current severity — schema gap, finding not fix

`scoring.yaml` says thresholds apply to *baseline* severity. The schema has only `current_severity`, which `compute_baseline_severity` writes into — they're the same value

today, so the test cannot distinguish them.

As soon as event ingestion applies `AxesImpact` deltas, `current_severity` will drift with the news, and tiers derived from it will drift with the news too.

Fix path (recorded in `_out/schema_gap.txt`, NOT applied in this task):

- Add `baseline_severity` alongside `current_severity` on `DynamicFields`.
- `refresh_all_derived` writes `baseline_severity`.
- `propagate_event` writes `current_severity` per-event, without touching tier.
- Tier derived from `baseline_severity` only.

## 7. Two things worth flagging about the tests themselves

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- `test_prose_percentages_match_edge_weights` currently passes, but its guard is fairly loose — it matches any edge weight *touching* the node. The strict invariant is “matches a weight of an edge in the specific bucket this prose sentence is about,” but that requires the narration builder to expose which edge produced which percentage. Kept the looser version to avoid coupling the test to narration internals. Flagged so nobody thinks this test is stronger than it is.
- `test_every_node_has_at_least_one_edge` passes — nothing is currently isolated. If a future data edit adds an unreferenced country or facility, this test catches it before the graph renders.

## 8. How to run

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```
# just the tests
cd backend && .venv/bin/python -m pytest -v tests/

# tests + bundled report artefacts (this document's inputs)
backend/.venv/bin/python -m backend.tests.run_report
```

Documented at the repo root in `README.md`.

## 9. What did not happen

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Per the “no fixes” clause of the spec: nothing was fixed. Not one edge added, not one weight changed, not one threshold moved, not one line of scoring / schema / config / narration touched.

The failures and the two probes are the deliverable.